JOSE EDUARDO ALVAREZ ROBLES

Mechatronics Engineer | Software & CAD Automation Specialist

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PROFESSIONAL SUMMARY

Bilingual Mechatronics Engineer with 4+ years of expertise in software development, CAD automation, and 3D parametric engineering. Proven track record at multinational manufacturing operations architecting DriveWorks/SolidWorks configurators, dynamic calculation engines, and live 3D visualizers. Versatile full-stack developer experienced in building custom enterprise ERP systems and leading large-scale operations. Licensed engineer (Cédula Profesional) fluent in English and Spanish.

TECHNICAL CORE COMPETENCIES

CAD & 3D Modeling: SolidWorks (Advanced), DriveWorks (Pro/Automation), Blender, NX, 3D Printing (Additive Mfg).

Automation & Logic: Parametric CAD Rules, Live Stress/Weight Calculations, Macro Scripting, System Integration.

Software & Web Dev: JavaScript (ES6+), HTML5, CSS3/SASS, Full-Stack Architecture, ERP Systems, MySQL/Databases.

Operations & Languages: Fleet Logistics, Fleet Route Optimization, Technical English (B2+/Advanced), Spanish (Native).

PROFESSIONAL EXPERIENCE

Software Development Engineer & CAD Automation Specialist — Dorner Latin America Jan 2022 – Jun 2025

- Architected and deployed complex interactive product configurators using **DriveWorks** and **SolidWorks** in a 100% English-speaking global engineering environment.
- Programmed custom logic engines for real-time mechanical calculations (stress, weight, load distribution) and automated pricing quotes.
- Engineered automated CAD macros that generated full-assembly production line models dynamically, reducing configuration timelines by over 70%.
- Implemented dynamic multi-language interface switches (English/Spanish) and integrated live 3D web previews for conveyor product lines.

Full-Stack Developer & Technical Consultant — Independent Projects 2023 – Present

- Designed and sold tailored web ERP solutions for local enterprises, including cold-storage inventory hubs with real-time entry/exit tracking and financial analytics.
- Developed an end-to-end management platform for an automotive service hub integrated with commercial databases (Compaq), featuring automated profit margin calculation, service yard tracking, and automated email workflows.
- Managed custom additive manufacturing (3D printing) projects from CAD modeling in Blender/SolidWorks to production and quality control.

Logistics & Operations Manager — Fleet Management 2021 – 2023

- Managed last-mile logistics operations across the Guadalajara Metro Area for high-volume accounts including **Amazon** (~1,000 daily packages) and Price Shoes.
- Supervised a team of 10+ drivers, overseeing optimal route scheduling, driver compliance, payroll execution, and fleet maintenance.

EDUCATION & CREDENTIALS

B.S. in Mechatronics Engineering (Licensed / Cédula Profesional) **Graduated: Jul 2017**
Universidad del Valle de Atemajac (UNIVA), Guadalajara